

AQUANT FIN V SERIES

A Layered Quantitative Market-State and Trade-Qualification Architecture for Intraday Systematic Trading

ALIM V19.2 PRO — Adaptive Liquidity & Intelligence Model

QUANTUM-2 V24.3 — Research, Execution-State & Audit Architecture

FORMAL RESEARCH MANUSCRIPT

Prepared for controlled technical and prospective-buyer evaluation. Pre-publication draft; not peer reviewed.
This edition intentionally excludes source code, credentials, complete proprietary Boolean logic, full parameter-interaction maps and unpublished research records.

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Abstract

This study presents AQUANT FIN V SERIES, a quantitative research and trading architecture developed to examine intraday trade qualification as a layered decision problem rather than as an isolated BUY/SELL signal. The principal implementation, ALIM (Adaptive Liquidity & Intelligence Model) V19.2 PRO, combines market-state variables, directional-flow proxies, statistical order-flow normalization, liquidity and structural context, pullback-quality analysis, exhaustion-aware veto logic, deterministic multi-factor scoring and adaptive position management. QUANTUM-2 V24.3 extends the architecture with explicit execution-state control, trade-level quantitative research telemetry, shadow auditing and controlled hypothesis governance.

The documented ALIM V19.2 PRO evaluation uses NIFTY 50 15-minute data over 3 November 2025 to 22 September 2026. Under the stated TradingView configuration, the historical strategy test produced 1,073 trades, 809 profitable trades (75.40%), a profit factor of 1.87, maximum drawdown of ₹21,553.75 (2.01%) and expectancy of ₹155.76 per trade. The reported P&L is a simulation under a fixed position-size normalization and is not presented as directly executable futures or options performance.

The evidence is interpreted as preliminary empirical evidence rather than proof of persistent live alpha. The research record also shows an architectural progression from ALIM V19.2 PRO as the core deterministic decision engine, through QUANTUM-2 as an explicit market-state and execution-state extension, and later R14 research revisions (see Glossary) incorporating qualification, MSET-B, runtime-profile classification, finite-state execution control and research auditing. These generations are treated as distinct research states rather than being merged into a single performance claim. In particular, an R14 source can represent CANONICAL-R14, LEGACY-R14 or CUSTOM behaviour depending on its governing producer toggles; therefore performance evidence must always be tied to an exact source revision and exact configuration. The principal unresolved questions remain out-of-sample robustness, walk-forward stability, parameter sensitivity, feature dependence, transaction costs, execution effects and regime dependence. The contribution is therefore framed as a quantitative decision architecture and a controlled research methodology with a defined validation path.

Keywords: quantitative trading; systematic trading; market-state modelling; liquidity; order-flow proxy; exhaustion; deterministic scoring; finite-state execution; intraday trading; NIFTY 50; quantitative research

Glossary of Abbreviations, IDs and Research Codes

This glossary defines the principal abbreviations, architecture terms and controlled research identifiers used in the manuscript. Internal research IDs are included to preserve traceability across experiments, audits and software revisions. They are not presented as standardized industry terminology.

A. System and Architecture Terms

Term / ID	Definition / Role
ALIM	Adaptive Liquidity & Intelligence Model.
AQUANT	AQUANT FIN V SERIES — the broader quantitative research and system architecture.
V19.2 PRO	ALIM V19.2 PRO — the principal demonstrated deterministic quantitative decision and risk-management engine.

QUANTUM-2	Research, execution-state and audit architecture extending the ALIM framework.
V24.3	QUANTUM-2 V24.3 baseline architecture.
R14	A QUANTUM-2 research/revision branch incorporating later qualification, execution and audit mechanisms.
FSM	Finite State Machine.
QRF	Quantitative Research Framework / trade-level research telemetry layer.
MSET-B	Market-state/trade-qualification module implemented in the R14 research branch.
TET-PRO	Trade/Execution Terminal research architecture associated with the V19.2/TET-PRO blueprint.
MBRL	Model/Bayesian Research Laboratory research environment.

B. Quantitative and Market Terms

Term / ID	Definition / Role
PF	Profit Factor.
P&L	Profit and Loss.
DD	Drawdown.
MFE	Maximum Favorable Excursion.
MAE	Maximum Adverse Excursion.
CI	Confidence Interval.
SD	Standard Deviation.
SE	Standard Error.
IS	In-Sample.
OOS	Out-of-Sample.
HTF	Higher Time Frame.
OF	Order Flow; where used in ALIM, directional-flow evidence is a price/volume proxy unless native transaction-level data are connected.
CVD	Cumulative Volume Delta.
OI	Open Interest.
PCR	Put-Call Ratio.
ATR	Average True Range.
ADX	Average Directional Index.
EMA	Exponential Moving Average.

VWAP	Volume-Weighted Average Price.
LR	Linear Regression.
ICP	Exhaustion/qualification score used by the system's exhaustion-control logic; it is not established as a calibrated probability.
NIFTY	NIFTY 50 index; the principal instrument represented in the documented V19.2 evaluation.

C. Research, Audit and Experiment IDs

Term / ID	Definition / Role
AUD-001	Audit finding concerning REST-gateway exception handling and FSM deadlock risk.
AUD-002	Audit/research experiment concerning hypothesis interaction and trade-selection evidence.
AUD-003	Audit finding concerning symmetric momentum cross-talk in H-031/H-032.
AUD-004	Audit finding concerning entropy semantic-field overwrite.
AUD-005	Audit finding concerning delta-Z outlier handling.
AUD-006	Audit finding concerning synthetic gamma / max-pain / dealer-positioning assumptions.
AQUANT-001-005	Additional AQUANT architecture/audit findings recorded during system development.
H028	Research hypothesis identifier used in the AUD-002 experimental series.
H031	Research hypothesis identifier used in the AUD-002 experimental series.
H032	Research hypothesis identifier associated with the momentum research branch.
H033	Research hypothesis identifier used in the H033 × H028 interaction experiment.
DS-001	Research dataset/filter identifier used in AUD-001.
R0	AUD-002 reference/no-evidence experimental condition.
R2	AUD-002 research revision/experimental condition.
R3	AUD-002 research revision/experimental condition.
EXP-QUANTUM-2	Research experiment identifier used by the QUANTUM-2 governance layer.

Research-ID convention. Identifiers such as H028, AUD-002, R2 and EXP-QUANTUM-2 are internal research and audit identifiers. They identify hypotheses, experiments, audit findings or controlled revisions and should not be interpreted as universally standardized financial terminology.

Terminology boundary. Some terms in this glossary are architecture-specific to the ALIM/AQUANT research framework rather than established industry standards. They are defined here and in the relevant methodological sections.

1. Introduction and Problem Formulation

AQUANT FIN V SERIES was developed around a practical research problem: an entry signal does not, by itself, fully represent the conditions under which a trade should be accepted, rejected, executed or managed. A technically attractive setup may occur during an unfavourable market state, after a liquidity event, during exhaustion, or under conditions in which the information supporting the setup deteriorates before execution.

The research response was to construct a layered architecture in which market observations are transformed into measurable features and contextual states; evidence is combined through deterministic qualification and veto logic; accepted positions are treated as lifecycles rather than isolated entries; and resulting trades are retained as research observations.

Research proposition

The central proposition is deliberately narrower than a claim of institutional prediction or machine intelligence: structured combinations of observable market information may provide more informative trade qualification and risk management than simpler signal-only decision processes. Whether this holds out-of-sample remains an empirical question.

2. Research Questions and Scope

The research is organized around questions that can be tested using the architecture and its telemetry. They are stated as research questions rather than as claims that have already been proven.

Research question	Current evidence status
Can layered market-state variables improve trade qualification?	Development experiments demonstrate measurable incremental changes in trade-selection outcomes relative to simpler reference conditions. AUD-002 produced materially different results across hypothesis combinations and reference conditions, including both positive and negative interactions. The evidence supports incremental information value at the development-sample level; independent out-of-sample validation is still required to establish robustness.
Does exhaustion-aware gating change the distribution of accepted trades?	The mechanism is implemented and operational in the tested architecture, and exhaustion-aware qualification affects whether otherwise qualifying setups are admitted or vetoed. However, a clean isolated ablation quantifying its independent contribution to profit factor, expectancy, drawdown and trade distribution has not yet been established.

Does explicit execution-state tracking add information beyond signal generation?	QUANTUM-2 demonstrably captures execution-state information beyond the analytical signal, including state transitions, conflicts, recoveries, latency, score deterioration and exit attribution. Its incremental predictive or decision value relative to signal-only execution remains to be quantitatively isolated.
Can trade-level MFE/MAE, duration and exit attribution improve position-management research?	Yes, the research telemetry materially expands the information available for position-management analysis. QRF records MFE, MAE, duration, exit reason, entry score and contextual variables, enabling conditional analysis beyond aggregate P&L and win rate. Whether using these measurements to alter management rules produces robust performance improvement remains to be tested.
Are observed effects stable across parameters and market regimes?	Not established. The research record demonstrates sensitivity to hypothesis combinations and configurations, including both positive and negative interactions. Formal parameter perturbation, regime segmentation, walk-forward testing and independent out-of-sample validation remain necessary to establish stability and persistence.

3. Research Contributions

- A layered market-state and trade-qualification architecture separating observation, mathematical transformation, statistical state, structural/liquidity interpretation, qualification, veto logic, execution state and position management.
- Integration of directional-flow proxies and statistical normalization with structural, liquidity and exhaustion information rather than reliance on a single indicator.
- An explicit deterministic execution-state architecture in QUANTUM-2 that distinguishes analytical signals from execution state and records conflicts, recoveries, latency and score deterioration.
- A trade-level Quantitative Research Framework (QRF) incorporating P&L, MFE, MAE, duration, exit reason, entry score and contextual variables.
- A controlled research methodology based on a frozen baseline, single-variable experiments, baseline comparison, retain/reject decisions and restoration of the baseline before subsequent experiments.

4. Conceptual Architecture

Layer	ALIM V19.2 PRO	QUANTUM-2 V24.3
Market state	VWAP, ATR, RSI, EMA, ADX	Expanded state classification
Flow	Candle-derived directional delta	Delta plus statistical z-score
Momentum	Breakout/momentum logic	Discrete velocity and acceleration
Liquidity	Sweep and related conditions	Internal/external liquidity states
Structure	Configurable structure gate	HH/HL/LH/LL state
Pullback	Non-exhausted pullback gate	Multi-class pullback quality
Exhaustion	ICP-style veto	Integrated exhaustion/state analysis
Decision	Deterministic weighted score	Score + gates + state context
Execution	Strategy lifecycle	Explicit finite-state execution
Research	Strategy statistics/HUD	QRF, MFE/MAE, attribution, CI

Audit	Limited	Shadow audit, conflicts, recovery, latency, score drift
Governance	Configuration-driven	Baseline/experiment/promotion framework

5. ALIM V19.2 PRO — Quantitative Architecture

5.1 Market-State Representation

The implementation uses VWAP, ATR, RSI, fast and slow EMA, volume statistics and ADX to establish directional, volatility and trend context. Higher-timeframe context is incorporated through a 45-minute filter in the documented validated configuration.

5.2 Directional-Flow Proxy

Buy- and sell-volume proxies are derived from the location of the closing price within the OHLC range and combined into a directional delta. This is an OHLC/volume-derived pressure proxy. It is not native exchange bid/ask transaction data and should not be described as direct observation of order-book participants.

5.3 Multi-Factor Qualification

The deterministic scoring layer combines configured contributions from bias, trend, higher-timeframe context, directional flow, PCR information where enabled, reversal structures, liquidity, divergence, session-open conditions, volume-cluster context, equilibrium, thrust, momentum and other active signals. Entry gates can additionally restrict trades according to session, trend strength, statistical-flow state, exhaustion, structure and pullback quality.

5.4 Exhaustion-Aware Qualification

The exhaustion component combines normalized impulse/range information, proximity to liquidity extremes and rejection characteristics followed by a bounded transformation. It is used as a qualification/veto mechanism. It is a model score, not a calibrated probability.

5.5 Adaptive Position Lifecycle

After a trade is accepted, position management can include dynamic stop logic, breakeven movement, partial profit taking, trailing protection, time-based exit and cooldown. This makes the research object a trade lifecycle rather than only an entry event.

6. QUANTUM-2 V24.3 — Research and Execution Extension

6.1 Discrete Dynamical Features

The architecture calculates a short-window linear-regression estimate and derives discrete velocity and acceleration:

$$v_t = LR_t - LR_{t-1} \quad ; \quad a_t = v_t - v_{t-1}$$

These are discrete-time first- and second-order changes. The present implementation should therefore not be described as a continuous differential-equation model.

6.2 Statistical Order-Flow State

Directional delta is standardized against its recent mean and standard deviation, producing a local z-score that measures unusual flow relative to the recent distribution.

6.3 Structure, Pullback and Liquidity State

Pivot-derived higher-high/lower-high/higher-low/lower-low information is maintained as structural context. Pullbacks are classified into states including absorption, healthy pullback, weak pullback, reversal attempt and reversal confirmed. Liquidity conditions include buy-side/sell-side liquidity taken and external liquidity targeting.

6.4 Multi-Timeframe Context

The architecture combines lower- and higher-timeframe state information so that a 15-minute decision is not interpreted solely from an isolated bar.

6.5 Architecture Evolution and Configuration Integrity

The research record indicates that AQUANT FIN V SERIES evolved through identifiable architectural generations rather than through simple accumulation of indicators. ALIM V19.2 PRO constitutes the principal deterministic decision and risk-management engine. QUANTUM-2 V24.3 extends that engine with explicit market-state representation, finite-state execution control, trade-level quantitative research telemetry and Shadow Audit logic. Subsequent R14 revisions add explicit qualification components such as MSET-B, runtime-profile classification and execution-routing controls. Separate research tooling, including the hypothesis-registry and Bayesian Research Laboratory work, belongs to the experimental research layer and should not be treated as evidence that every hypothesis displayed by a research interface has been independently validated or promoted to production. R14, MSET-B and MBRL are internal AQUANT architecture/research identifiers; their definitions are provided in the Glossary of Abbreviations, IDs and Research Codes.

A configuration-integrity rule follows from this lineage: a named version is not, by itself, a sufficient identifier of a tested strategy. The R14 implementation explicitly distinguishes CANONICAL-R14, LEGACY-R14 and CUSTOM runtime profiles using six producer-selector toggles, while other feature-enable and policy settings remain orthogonal configuration dimensions. Consequently, empirical claims should be reported as source revision + runtime profile + governing feature/policy configuration + data period + test environment. This prevents results from different generations or configurations from being implicitly pooled into one undifferentiated performance record.

6.6 Game-Theoretic Strategic Interaction Layer

The broader AQUANT research history also included a Strategic Interaction Engine (SIE) exploring game-theoretic concepts for participant classification and strategic interaction. The research layer incorporated participant types, strategy weights, payoff relationships, best-response-style analysis and Nash-stability measures. In its documented implementation, these were heuristic and proxy-based constructs rather than a formal Nash-equilibrium, Stackelberg, mean-field, or evolutionary-game solver. Accordingly, the SIE outputs are interpreted as strategic-interaction indicators rather than mathematically solved market equilibria.

7. Deterministic Execution-State Architecture

A central QUANTUM-2 extension is the explicit separation of analytical signal generation from execution state. The finite-state architecture includes WAITING, ENTRY PENDING, LONG, SHORT, EXIT PENDING, REVERSE PENDING and STATE CONFLICT states, together with recovery handling.

The Shadow Audit records fresh and reversal signals, accepted and rejected executions, timeouts, conflicts, recoveries, signal-to-execution latency and score drift. This creates a researchable distinction between theoretical signal quality and realized execution state.

8. Quantitative Research Framework

Observation	Research use
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P&L	Outcome magnitude
MFE	Maximum favourable excursion and exit efficiency
MAE	Maximum adverse excursion and stop quality
Duration	Time-in-trade behaviour
Exit reason	Outcome attribution
Entry score	Score/outcome relationship
ATR / ADX	Regime conditioning
VWAP distance	Location/context
AM / PM	Intraday segmentation
Mean / median / SD	Distributional description
SE / CI	Uncertainty around sample estimates

The implemented confidence interval is a simple normal approximation around the sample mean. It is not treated as a complete inferential treatment of serial dependence, heteroskedasticity or regime shifts.

9. Research Methodology and Governance

Stage	Research action
1. Frozen baseline	Preserve a stable reference configuration.
2. Hypothesis	Define one falsifiable research proposition.
3. Mathematical feature	Translate the proposition into measurable variables.
4. Single-variable experiment	Change one research element at a time.
5. Baseline comparison	Measure incremental effect.
6. Qualification / veto test	Evaluate trade-selection impact.
7. Execution analysis	Evaluate lifecycle and execution behaviour.
8. Telemetry / audit	Capture trade-level and state-level evidence.
9. Retain / reject	Apply predefined research criteria.
10. Restore baseline	Return to the frozen reference.
11. Independent validation	Test robustness on unseen data.

The purpose of this protocol is to reduce uncontrolled feature accumulation and make changes attributable to explicit research hypotheses. A module is not promoted merely because it improves one backtest statistic.

Version-controlled research lineage is therefore treated as part of the methodology itself. A change that alters a producer, qualifier, veto, execution rule or risk-management rule creates a new experimental condition and should not be compared with the frozen baseline without recording the exact configuration. This is especially important where later revisions contain both canonical and legacy paths.

10. Preliminary Empirical Evaluation

The documented performance evaluation concerns ALIM V19.2 PRO on NIFTY 50 15-minute data from 3 November 2025 through 22 September 2026.

Metric	Observed / reported value
Total trades	1,073
Profitable trades	809 / 1,073
Profitable-trade rate	75.40%
Profit factor	1.87
Maximum drawdown	₹21,553.75 / 2.01%
Expectancy	₹155.76 per trade

Gross profit	₹359,272.20
Gross loss	₹192,138.15
Position-size normalization	25 NIFTY units

The TradingView historical test used a fixed 25-unit NIFTY position-size normalization. The NIFTY 50 index itself is not a directly tradable contract; therefore the reported P&L is not presented as directly executable futures or options performance. Instrument-specific implementation requires separate validation. The 25-unit normalization is used solely as a consistent scaling reference across the test period; it does not represent a specific futures lot size or a deployable position. Accordingly, the reported rupee P&L is a simulation under that normalization and should not be translated into actual futures or options P&L without separate instrument-specific validation.

Configuration-specific evidence boundary

The reported 1,073-trade result belongs to the documented ALIM V19.2 PRO TradingView configuration described in this manuscript. It should not be retroactively attributed to QUANTUM-2 V24.3, R14, MBRL or any later research revision unless that exact configuration has been independently run and its results recorded. Likewise, historical variants observed during development may exhibit different trade counts, profit factors or drawdown characteristics, but no such variant is included in the present empirical table unless its exact source revision, configuration and reproducible test record are available. This rule deliberately favours traceability over selective performance reporting.

Interpretive boundary

The historical result demonstrates that the documented configuration is empirically testable and produced a positive historical strategy distribution under the stated simulation conditions. It does not establish persistent live alpha, future profitability, broker-level execution quality or robustness outside the tested period.

11. Research Integrity and Evidence Boundaries

Claim area	Defensible interpretation
Order flow	Directional-flow proxy unless native order/transaction data are connected.
AI / ML	Deterministic quantitative scoring; no trained ML model is established by the reviewed implementation.
Institutional behaviour	Observable institutional-style proxies; no direct participant identification.
Probability	Model score unless independently calibrated.
Differential dynamics	Discrete velocity/acceleration; continuous model remains future research.
Game theory	Conceptual research direction; no formal equilibrium solver.
Execution	TradingView strategy execution model unless broker-validated.
Performance	Historical/backtested result under the stated configuration.

These boundaries are intentional. Precise description of implemented capability versus future research is treated as part of the research integrity of the project.

The same boundary applies to terminology. Directional flow in the reviewed implementation is a candle-derived volume proxy rather than native bid/ask transaction data; the displayed AI score is a deterministic weighted quantitative score rather than a trained machine-learning model; the exhaustion/ICP mechanism is a model score rather than a calibrated probability; and velocity/acceleration are discrete differences derived from linear-regression estimates rather than a continuous differential-equation model. These distinctions are preserved because they affect how the architecture can be independently evaluated.

12. Threats to Validity and Limitations

Research-infrastructure versus validated-result boundary

The later research environment contains hypothesis registries, Bayesian-style posterior displays, promotion ladders and information-oriented diagnostics. These components demonstrate an experimental framework for organising hypotheses and observations; they do not, by themselves, establish statistical significance, causal validity or production readiness for the hypotheses displayed. Promotion must remain conditional on predefined sample requirements, independent testing, multiple-comparison awareness, out-of-sample evidence and restoration of the frozen baseline when a hypothesis fails.

- Overfitting and parameter-selection risk remain material because the architecture has a substantial parameter surface.
- Trend, EMA, VWAP, higher-timeframe alignment, ADX, structure and momentum can contain correlated information; incremental information must be tested.
- TradingView strategy results do not establish broker fills, slippage, latency or market impact.
- The directional-flow proxy inherits the limitations of OHLC and total-volume data.
- Performance may vary across trend, volatility, event and liquidity regimes.
- NIFTY index results cannot be assumed to transfer unchanged to futures or options.
- Simple confidence intervals do not fully address dependence or regime shifts.
- The OI-divergence component in the reviewed source is a placeholder and should not be represented as an active independent OI signal.
- The current architecture does not constitute a formal machine-learning, game-theoretic equilibrium or continuous differential-equation model.

13. Market-Microstructure Interpretation

The research can be interpreted through a market-microstructure lens because it uses observable price, volume, flow and liquidity behaviour as imperfect proxies for heterogeneous market participants. Existing research has documented relationships between order flow, information and price formation, including evidence that order flow can contain information relevant to market prices and that strategic traders may operate in informationally complex environments [1–4].

The relevance of this literature is contextual rather than evidentiary for AQUANT. The cited studies do not validate the AQUANT architecture, its NIFTY results or its particular proxies. They provide theoretical and empirical context for why flow and participant heterogeneity can be legitimate objects of quantitative investigation.

Accordingly, the system does not claim to directly observe institutions, hidden orders or strategic equilibria. Its contribution is the operationalization and combination of observable variables into a deterministic qualification and execution architecture.

14. Academic Value and Research Contribution

14.1 Methodological contribution

The principal academic value is methodological. The framework treats signal generation, qualification, veto logic, execution state and post-trade measurement as separate research objects. This creates a basis for testing incremental information rather than attributing the entire outcome to a single indicator.

14.2 Contribution to systematic-trading research

The architecture provides an experimental bridge between feature engineering and execution research. By explicitly recording signal state, execution state, latency, conflicts, recovery and trade-level excursions, it permits investigation of whether implementation-state variables contain information about realized outcomes.

14.3 Contribution to market-state research

Market state is represented through multiple observable dimensions—trend, volatility, directional-flow proxy, structure, liquidity, pullback quality and exhaustion. The research question is not whether these variables are individually novel, but whether their conditional combination and sequencing provides incremental decision value.

14.4 Contribution to exhaustion research

The exhaustion-veto concept creates a falsifiable program: compare a qualified-signal baseline with a matched architecture that additionally rejects exhausted conditions. Outcomes should include expectancy, profit factor, drawdown, MFE, MAE, duration, missed-winner rate and out-of-sample stability.

14.5 Contribution to execution research

The QUANTUM-2 finite-state architecture makes execution-state deterioration observable. Signal-time score and execution-time score can be compared, and latency can be grouped against outcomes. This creates a measurable question concerning whether trade quality decays between analytical recognition and execution.

14.6 Contribution to research governance

The frozen-baseline and single-variable methodology is intended to reduce uncontrolled model evolution. It provides a practical mechanism for preserving experimental lineage, restoring a reference state and evaluating hypotheses independently before considering multi-factor promotion.

14.7 Falsifiability

The framework is deliberately falsifiable. A proposed module can be rejected if it fails to improve predefined measures, improves only in-sample performance, is unstable under parameter perturbation, or loses its effect after realistic costs. This distinction separates a research architecture from a promotional trading claim.

14.8 Version lineage and configuration traceability

A further methodological contribution is the explicit treatment of software lineage and runtime configuration as empirical variables. In a complex systematic strategy, two source revisions carrying similar feature names may implement materially different qualification or execution behaviour. By recording the frozen baseline, source revision, runtime profile, policy selectors and feature enables alongside the result, the research process makes configuration drift observable and reduces the risk of attributing a result to the wrong algorithmic generation. This is a software-research governance contribution rather than a claim of statistical performance improvement.

15. Intellectual Architecture and Practical Relevance

Individual components such as EMA, RSI, ATR, VWAP, liquidity sweeps, CVD or statistical normalization are not claimed as proprietary discoveries. The research asset lies in the specific architecture: the combination and sequencing of evidence, weighting and thresholds, exhaustion-aware gating, structural and liquidity context, adaptive position lifecycle, deterministic execution-state control, trade-level telemetry and controlled hypothesis promotion.

Central architectural proposition

The visible BUY/SELL marker is an output layer. The more important research object is the decision architecture that determines when a signal is permitted to become a trade, how that trade is managed, and how the resulting observation feeds subsequent research.

16. Controlled Validation Roadmap

16.1 Historical-version reconciliation

Where earlier development versions show materially different trade-frequency or profit-factor profiles, the appropriate research test is not to select the most attractive historical statistic. Instead, each candidate must be reconstructed from its exact source revision and configuration, evaluated over the same data protocol, and compared using predefined criteria including trade count, expectancy, drawdown, profit factor, distribution of returns, parameter sensitivity and out-of-sample behaviour. Only then can an observed difference be interpreted as an architectural effect rather than configuration or selection bias.

Research test	Primary question	Required evidence
Independent reproduction	Can an independent evaluator reproduce the documented result?	Same configuration, data assumptions and reported statistics.
Strict OOS	Does performance survive unseen observations?	Predefined holdout with no parameter re-selection.
Walk-forward	Does the architecture remain stable through sequential re-estimation?	Multiple rolling training/testing windows.
Ablation	Which modules add incremental information?	Matched baseline and module-by-module removal.
Parameter perturbation	Is performance localized to narrow parameter values?	Sensitivity surfaces and perturbation ranges.
Cost sensitivity	Does the effect survive realistic friction?	Slippage, fees and execution assumptions.
Regime analysis	Does behaviour change across market states?	Trend/volatility/event/liquidity segmentation.
Forward execution	Does the strategy behave similarly outside historical simulation?	Paper or controlled live execution with logs.
Instrument transfer	Does the logic transfer to futures/options only after instrument-specific modelling?	Separate instrument-level validation.

16.2 Cross-Experiment Evidence: Positive, Negative and Non-Promoted Results

A cross-check of the AQUANT research record identifies a second evidence layer beyond the current 1,073-trade ALIM V19.2 PRO evaluation: controlled hypothesis and audit experiments in which candidate mechanisms were compared against baseline or TAKE/REJECT reference conditions. These records are treated separately from the current V19.2 result because they belong to earlier research experiments and often have much smaller sample sizes. Their principal value is therefore hypothesis discrimination, not performance marketing.

One documented audit series provides a useful example of why negative and null results matter. AUD-001 reported that a $PF > 1$ condition was achieved on 5 of 6 evaluated cases, while TAKE conditions exceeded REJECT conditions on 5 of 6 cases; SENSEX was an exception in which the ordering was reversed. The associated research interpretation classified DS-001 as a weak-positive filter rather than a strong classifier. This result should not be generalized across instruments, but it demonstrates that the research process retained an instrument-level exception rather than treating the aggregate result as universal.

A second documented audit series (AUD-002) produced a more discriminating result. The H028×H031 R2 combination produced PF 1.90 for NIFTY (n=19) and PF 2.50 for BANKNIFTY (n=20), compared with TAKE conditions of PF 1.26 and 1.24 respectively. When H028 was killed, aggregate PF values fell to 1.41 and 1.31. By contrast, an R3 H033×H028 combination produced PF 0.98 for NIFTY and 0.37 for BANKNIFTY. An R0 no-evidence condition remained competitive at PF 1.50 (n=44) for NIFTY and 1.24 (n=46) for BANKNIFTY. These results are development-sample evidence and have small samples; they are therefore interpreted as hypothesis-ranking evidence rather than validated alpha. Nevertheless, they demonstrate an important

research property: adding a hypothesis did not automatically improve the outcome, and some combinations were materially worse than simpler reference conditions. These sample sizes are too small to support reliable conventional statistical inference on performance differences; the observed PF differences may be compatible with sampling noise.

The most important academic implication is that AQUANT's research history contains both positive and negative selection evidence. A candidate mechanism can be retained, rejected, classified as weakly positive, or left unresolved. This is materially different from reporting only the final successful configuration. The commercial implication is also relevant: a prospective technical evaluator can examine not only what was retained but also which candidate mechanisms failed, were unstable across instruments, or failed to improve a simpler reference.

The cross-experiment figures above are not pooled with the 1,073-trade ALIM V19.2 PRO result. They arise from separate research/audit samples, instruments and hypothesis conditions. They are included because they document the selection process and failure modes, not because they establish a common performance track record. Independent replication, larger samples, out-of-sample testing and multiple-comparison controls remain necessary before these findings can be promoted to general performance claims.

17. Conclusion

AQUANT FIN V SERIES represents an attempt to formalize intraday systematic trading as a layered quantitative decision problem. ALIM V19.2 PRO is the principal demonstrated strategy implementation; QUANTUM-2 V24.3 extends the architecture toward explicit execution-state control, quantitative trade telemetry, shadow auditing and research governance.

The documented 1,073-trade NIFTY 50 15-minute historical evaluation provides substantial preliminary evidence for continued research, but it is not sufficient to establish out-of-sample robustness or persistent live performance. The next scientific stage is therefore stronger validation rather than uncontrolled complexity.

The research program is strongest when it maintains a strict distinction between implemented functionality, demonstrated historical evidence and future hypotheses. Independent reproduction, strict out-of-sample testing, walk-forward analysis, parameter perturbation, ablation, transaction-cost sensitivity, regime analysis and forward execution validation are the appropriate next gates.

18. References

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Only the five external works listed above are used as external literature in this manuscript. No additional references, empirical findings or literature claims have been added beyond the sources used for the research context.

Appendix A. Controlled Disclosure and Reproducibility Boundary

Disclosure stage	Information available
This manuscript	Research problem, architecture, methodology, empirical evidence, limitations and validation roadmap.
Technical discussion	Architecture walkthrough, configuration rationale and selected evidence.
NDA diligence	Deeper implementation description, controlled reproducibility and detailed research records.
Qualified technical diligence	Potential source-level access under agreed terms and controls.
Transaction stage	IP inventory, transfer/licence scope, documentation, support and knowledge transfer.

The manuscript is intentionally sufficient for a technically informed reader to understand the research proposition, architecture, evidence and limitations without disclosing the implementation details that constitute the project's protected IP.

Appendix A.1. Configuration traceability rule

For future empirical releases, every reported result should carry a minimum traceability tuple: (1) source-code revision, (2) runtime profile, (3) feature and policy configuration, (4) instrument and timeframe, (5) date range, (6) position-size normalization, and (7) execution/backtest environment. A result without this tuple is treated as development evidence rather than a formal empirical result.

Appendix B. Author Statement

I developed the AQUANT FIN V SERIES research architecture iteratively as an independent quantitative research project. Pine Script and AI-assisted engineering tools were used during implementation, debugging, refactoring and technical analysis. The research hypotheses, architecture, configurations, testing requirements and acceptance criteria were defined and evaluated as part of the research process. This manuscript is intended as a transparent account of the current research state rather than a claim that the system has completed institutional-grade independent validation.

Manuscript status

Formal research draft for controlled technical and prospective-buyer evaluation. Pre-publication draft; not peer reviewed, independently audited, or investment advice.

Appendix C. Research Contribution Matrix

Research dimension	Contribution made by framework	What remains to be demonstrated
Trade qualification	Layered state + score + veto architecture	Incremental value through ablation/OOS
Exhaustion	Explicit exhaustion-aware rejection mechanism	Robustness and missed-opportunity analysis
Execution	Finite-state signal/execution separation	Live/forward execution validation
Telemetry	MFE/MAE, latency, score drift, exit attribution	Statistical inference under dependence
Research governance	Frozen baseline and single-variable	Independent replication

	experiments	
Market microstructure	Observable liquidity/flow proxies	Validation with richer transaction/order-book data
Mathematical dynamics	Discrete velocity/acceleration	Whether dynamic features add stable predictive value
Empirical evidence	1,073-trade preliminary study	Strict OOS, walk-forward, costs and regimes

Appendix D. Experimental Evidence Register and Interpretation Rules

A further rule is applied to negative results: an unsuccessful experiment is retained as evidence only when the experimental condition and comparison are identifiable. Where an earlier record indicates that an experiment occurred but the numerical result cannot be reconstructed reliably, the record is not converted into a numerical finding. This protects the manuscript against retrospective reconstruction and selective reporting.

The experimental register separates four evidence classes: (A) demonstrated historical performance under a fully specified configuration; (B) development-sample hypothesis evidence; (C) qualitative/code-level architectural evidence; and (D) unresolved or future hypotheses. Only class A is used for the principal V19.2 performance table. Class B is used to document research discrimination and model-selection behaviour. Class C supports architectural claims but not statistical performance claims. Class D is explicitly excluded from empirical conclusions.

Evidence ID	Research condition	Observed evidence	Sample	Interpretation	Status
AUD-001	DS-001 / TAKE vs REJECT	PF > 1 in 5/6; TAKE > REJECT in 5/6; SENSEX reversed	6 cases	Weak positive filter; instrument-specific exception	Finding closed
AUD-002 R2	H028 × H031	PF 1.90 NIFTY; PF 2.50 BANKNIFTY vs TAKE 1.26 / 1.24	n=19 / 20	Strong candidate within development sample	Replication required
AUD-002 killed	H028 removed	Aggregate PF 1.41 NIFTY / 1.31 BANKNIFTY	Development sample	Suggests H028 contributed incremental value in that test	Replication required
AUD-002 R3	H033 × H028	PF 0.98 NIFTY / 0.37 BANKNIFTY	Development sample	Negative interaction; not evidence for promotion	Not promoted
AUD-002 R0	No-evidence reference	PF 1.50 NIFTY / 1.24 BANKNIFTY	n=44 / 46	Competitive simple reference; anti-overfitting control	Replication required